Brownian Motion, Stopping Times, and Quadratic Variation

References

Main references

- S. Shreve, *Stochastic Calculus for Finance II*, Chapter 3.
- I. Karatzas and S. Shreve, *Brownian Motion and Stochastic Calculus*, Chapter 2.
- T. Björk, *Arbitrage Theory in Continuous Time*, chapters on Brownian motion and stochastic integration.

Roadmap

1. Define continuous-time stochastic processes and filtrations.
2. Construct Brownian motion and identify its Gaussian structure.
3. Read the weak Markov property as renewal at deterministic times.
4. Introduce stopping times, $\mathcal{F}_\tau$, and strong Markov renewal.
5. Organize the reflection principle as a path transformation.
6. Measure path roughness through quadratic variation.
7. Turn Brownian motion into a positive price model.

Continuous-Time Stochastic Processes

A stochastic process is a time-indexed family of random variables

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let E be a state space.

Stochastic process

A continuous-time stochastic process is a family

$$X = (X_t)_{t \geq 0}, \qquad X_t : \Omega \to E,$$

such that each X_t is a random variable.

Time parameter

- discrete: $t \in \{0, 1, 2, \ldots\}$;
- continuous: $t \in [0, \infty)$.

Typical state spaces

- $E = \mathbb{R}$: price or factor;
- $E = \mathbb{R}^d$: several risk factors;
- $E = \mathbb{N}$: event count.

One process has both distributional and pathwise descriptions

For $X : [0, \infty) \times \Omega \to E$, we can freeze either argument.

Random variable at time t

Fix t. Then $X_t : \omega \mapsto X_t(\omega)$ describes uncertainty across scenarios at that date.

The joint laws

$$(X_{t_1}, \ldots, X_{t_n})$$

are the finite-dimensional distributions.

Sample path in scenario ω

Fix ω. Then

$$t \longmapsto X_t(\omega)$$

is one realized trajectory through time.

Path-dependent claims depend on the whole trajectory, not only on X_T.

Important

Marginal laws alone do not determine a process: dependence across dates matters.

Path regularity and increments distinguish continuous-time models

Continuous paths

The process has continuous paths if

$$\mathbb{P}\big(t \mapsto X_t \text{ is continuous}\big) = 1.$$

Càdlàg paths

Paths are right-continuous and have left limits. This allows jumps while ruling out pathological oscillations.

For $0 \leq s < t$, the increment is

$$X_t - X_s.$$

- **stationary increments:** its law depends only on $t - s$;
- **independent increments:** increments over disjoint intervals are independent.

Brownian motion has continuous paths and both increment properties.

A filtration records how information grows through time

Filtration

A filtration $(\mathcal{F}_t)_{t \geq 0}$ is an increasing family of sigma-fields:

$$\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}, \qquad 0 \leq s \leq t.$$

The sigma-field $\mathcal{F}_t$ represents everything observable by time t.

Natural filtration of a process

$$\mathcal{F}_t^X = \sigma(X_u : 0 \leq u \leq t).$$

It contains exactly the information revealed by the path of X up to time t.

In continuous time, one usually augments this filtration by null events and imposes right-continuity; these are the *usual conditions*.

Adaptedness prevents information from arriving from the future

Adapted process

A process X is adapted to $(\mathcal{F}_t)$ if X_t is $\mathcal{F}_t$-measurable for every $t \geq 0$.

At time t, an adapted quantity can be computed from currently available information.

Admissible

With $\mathcal{F}_t^W = \sigma(W_u : u \leq t)$, both W_t and $\max_{0 \leq u \leq t} W_u$ are $\mathcal{F}_t^W$-measurable.

Not observable yet

For $h > 0$, neither W_{t+h} nor the sign of $W_{t+h} - W_t$ is known at time t.

In finance, a trading decision must be adapted – and, for stochastic integration, typically predictable.

Three conditional notions organize the rest of the lecture

For an adapted process X and $0 \leq s < t$:

Martingale: if X_t is integrable and

$$\mathbb{E}[X_t \mid \mathcal{F}_s] = X_s,$$

the current value is the best conditional mean forecast.

Markov process: the conditional law of the future given $\mathcal{F}_s$ depends on the past only through the present state X_s.

Stopping time: a random time τ satisfies

$$\{\tau \leq t\} \in \mathcal{F}_t;$$

whether stopping has occurred is observable without looking ahead.

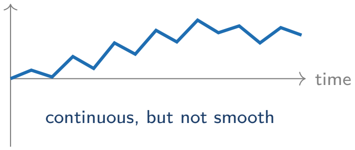
Brownian motion provides the canonical example of all three ideas.

Brownian Construction and Gaussian Structure

A useful continuous-time model must keep randomness and roughness

Financial prices receive many small shocks:

- information arrives through time;
- the direction of the next move is uncertain;
- paths look continuous at a daily scale but irregular when we zoom in.



Modeling goal

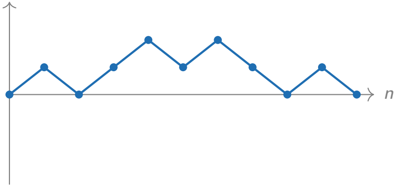
Construct a continuous process whose increments remain random at every time scale.

The simple random walk accumulates independent shocks

Let $\varepsilon_1, \varepsilon_2, \ldots$ be i.i.d. with

$$\mathbb{P}(\varepsilon_k = 1) = \mathbb{P}(\varepsilon_k = -1) = \tfrac{1}{2}, \qquad S_n = \sum_{k=1}^{n} \varepsilon_k.$$

- S_n is the position after n steps.
- $\mathbb{E}[S_n] = 0$ and $\mathrm{Var}(S_n) = n$.
- A typical displacement therefore has size $\sqrt{n}$.



Rescaling balances more steps with smaller shocks

To fit n shocks into one unit of time, shrink each shock by $1/\sqrt{n}$:

$$W_t^{(n)} = \frac{S_{\lfloor nt \rfloor}}{\sqrt{n}}, \qquad 0 \le t \le 1,$$

with linear interpolation between grid points.

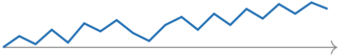
$$n \text{ shocks} \longrightarrow \text{size } 1/\sqrt{n} \longrightarrow \mathrm{Var}(W_1^{(n)}) = 1$$

Why $1/\sqrt{n}$?

The total variance remains of order one: $n \times (1/\sqrt{n})^2 = 1$.

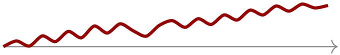
Donsker explains why Brownian motion is the natural limit

Rescaled random walk



$\Longrightarrow$

Brownian path



Donsker's invariance principle, informal

As the mesh goes to zero, the rescaled random-walk path $W^{(n)}$ converges in distribution to a standard Brownian motion W.

The theorem motivates the model; its technical proof is not needed for what follows.

Four properties define standard Brownian motion

Standard Brownian motion

A process $(W_t)_{t\geq 0}$ is a standard Brownian motion if:

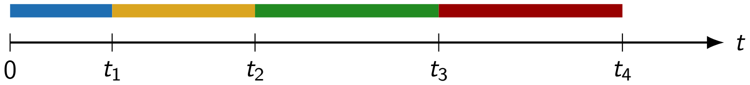
1. $W_0 = 0$;
2. increments over disjoint time intervals are independent;
3. $W_t - W_s \sim \mathcal{N}(0, t-s)$ for $0 \leq s < t$;
4. $t \mapsto W_t$ is continuous almost surely.

One-line interpretation

Fresh shocks arrive independently, their scale depends only on elapsed time, and their accumulation produces a continuous but rough path.

Independent increments mean that non-overlapping shocks are unrelated

$$\Delta W_i = W_{t_i} - W_{t_{i-1}}$$



For $0 = t_0 < t_1 < \cdots < t_m$, the increments

$$W_{t_1} - W_{t_0}, \ldots, W_{t_m} - W_{t_{m-1}}$$

are independent.

Concrete meaning

Knowing that the path rose between t_1 and t_2 does not change the distribution of its move between t_3 and t_4.

Stationary Gaussian increments turn elapsed time into uncertainty

For every $0 \leq s < t$,

$$W_t - W_s \sim \mathcal{N}(0, t - s).$$

Stationary increments

The law depends on the duration $t - s$, not on the calendar location s.

Gaussian increments

Every increment is normally distributed, centered, and symmetric.

$$W_{1.5} - W_{0.5} \overset{d}{=} W_{11} - W_{10} \sim \mathcal{N}(0, 1).$$

A four-times-longer interval produces twice the typical displacement.

$$\mathrm{SD}(W_t - W_s) = \sqrt{t - s}.$$

Small examples: use the four Brownian properties immediately

Let $(W_t)_{t\geq 0}$ be a standard Brownian motion with its natural filtration.

1. Since only elapsed time matters,

$$W_4 - W_1 \sim \mathcal{N}(0, 3).$$

2. Since $W_1 \in \mathcal{F}_2$ and $W_4 - W_2$ is independent of $\mathcal{F}_2$,

$$\mathrm{Cov}(W_1, W_4 - W_2) = 0.$$

3. By symmetry of every fresh Gaussian increment, for $h > 0$,

$$\mathbb{P}(W_{t+h} > W_t \mid \mathcal{F}_t) = \tfrac{1}{2}.$$

Each calculation uses a different ingredient: stationarity, independence, or Gaussian symmetry.

Quick diagnostic: which transformations are Brownian motions?

Starting from a standard Brownian motion W, consider

$$X_t = -W_t, \qquad Y_t = 2W_{t/4}, \qquad Z_t = W_{1+t}.$$

Your turn – 90 seconds

For each process, check:

1. its value at $t = 0$;
2. independence of increments;
3. increment variance;
4. path continuity.

Conclusion

- X is Brownian by symmetry.
- Y is Brownian by scaling: $4\,\mathrm{Var}(W_{t/4}) = t$.
- Z is not standard Brownian because $Z_0 = W_1 \neq 0$. The corrected process $W_{1+t} - W_1$ is Brownian.

Gaussian vectors reduce joint laws to linear algebra

Gaussian vector

A random vector $Z = (Z_1, \dots, Z_m)^\top$ is Gaussian if every linear combination

$$a^\top Z = \sum_{i=1}^{m} a_i Z_i, \qquad a \in \mathbb{R}^m,$$

is a one-dimensional Gaussian random variable.

Fundamental characterization

Its law is completely determined by its mean vector $m_Z = \mathbb{E}[Z]$ and covariance matrix $\Sigma_Z = (\mathrm{Cov}(Z_i, Z_j))_{i,j}$.

In particular, a centered Gaussian vector is determined by its covariance matrix.

Brownian motion is a centered Gaussian process

Gaussian process

A process $(X_t)_{t\geq 0}$ is Gaussian if $(X_{t_1},\dots,X_{t_m})$ is a Gaussian vector for every finite set of times.

For $0 = t_0 < t_1 < \cdots < t_m$, write

$$W_{t_k} = \sum_{j=1}^{k} \big(W_{t_j} - W_{t_{j-1}}\big).$$

The increment vector has independent Gaussian coordinates. The level vector is a linear transformation of it, hence is Gaussian.

Conclusion

Brownian motion is a centered Gaussian process. Its finite-dimensional laws are therefore determined by its covariance kernel.

The Brownian covariance kernel is shared history

For $0 \le s \le t$, decompose

$$W_t = W_s + (W_t - W_s),$$

where the new increment is independent of W_s. Hence

$$K(s,t) := \mathrm{Cov}(W_s, W_t) = s = \min(s,t).$$

For instance, at times $1, 2, 4$,

$$\mathrm{Cov}\big((W_1, W_2, W_4)^\top\big) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 4 \end{pmatrix}.$$

Why call K a kernel?

For any dates and coefficients, $\sum_{i,j} a_i a_j K(t_i, t_j) = \mathrm{Var}(\sum_i a_i W_{t_i}) \ge 0$. Thus K generates valid covariance matrices and, with zero mean, every finite-dimensional Brownian law.

Brownian transformations give reusable processes

Symmetry, scaling, and time shifts

For $c > 0$ and $s > 0$, the processes

$$(-W_t)_{t\geq 0} \quad \text{and} \quad \left(\frac{W_{ct}}{\sqrt{c}}\right)_{t\geq 0} \quad \text{and} \quad (W_{s+t} - W_s)_{t\geq 0}$$

are standard Brownian motions. Moreover,

$$(W_{s-t} - W_s)_{0\leq t\leq s}$$

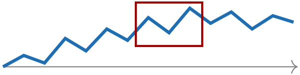
has the law of a Brownian motion restricted to $[0, s]$.

Symmetry powers reflection; scaling gives the square-root-of-time rule; time shifts express renewal and time reversal.

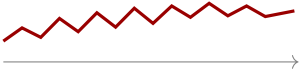
Proof idea: check the four defining properties and compute each increment variance.

Continuity hides infinite local roughness

A path segment



Zoomed segment



Continuity means no jumps.

Path roughness

Almost surely, a Brownian path is nowhere differentiable and has infinite total variation on every nontrivial interval.

Heuristic: the difference quotient has variance $\mathrm{Var}((W_{t+h} - W_t)/h) = 1/h$, which explodes as $h \downarrow 0$.

Bridge to quadratic variation

Ordinary variation explodes, yet squared increments have a finite accumulated limit: $[W]_t = t$. This is the roughness scale needed by stochastic calculus.

Liminf and limsup quantify Brownian oscillations

Perpetual oscillation

Almost surely,

$$\limsup_{t\to\infty} W_t = +\infty, \qquad \liminf_{t\to\infty} W_t = -\infty.$$

Brownian motion therefore crosses every fixed level, positive or negative. Nevertheless, on the linear scale,

$$\liminf_{t\to+\infty} \frac{W_t}{t} = \limsup_{t\to+\infty} \frac{W_t}{t} = 0.$$

For a two-sided Brownian motion, the same identities hold as $t \to -\infty$.

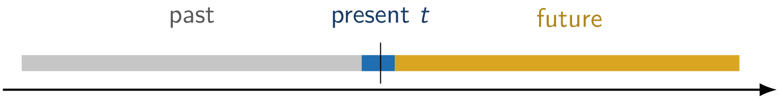
Law of the iterated logarithm

$$\limsup_{t\to\infty} \frac{W_t}{\sqrt{2t\log\log t}} = 1, \qquad \liminf_{t\to\infty} \frac{W_t}{\sqrt{2t\log\log t}} = -1,$$

$$\limsup_{t\downarrow 0} \frac{W_t}{\sqrt{2t\log\log(1/t)}} = 1, \qquad \liminf_{t\downarrow 0} \frac{W_t}{\sqrt{2t\log\log(1/t)}} = -1.$$

Markov Property as Renewal

The Markov idea separates past, present, and future



Markov principle, informal

Given the present state, additional knowledge of the past does not improve the conditional law of the future.

The future is not independent of the present. The present is precisely the state that links past and future.

At a fixed time, Brownian motion renews itself

Fix $t \geq 0$ and define the future increments

$$\widetilde{W}_u = W_{t+u} - W_t, \qquad u \geq 0.$$

Deterministic-time renewal

$(\widetilde{W}_u)_{u \geq 0}$ is a standard Brownian motion independent of $\mathcal{F}_t$.

Proof idea: $\widetilde{W}_0 = 0$; its increments are Brownian increments after time t, so they are independent of the past, Gaussian with variance equal to elapsed time, and stationary. Continuity is inherited from W.

Interpretation

At time t, retain the current level W_t and restart the noise from zero.

The weak Markov property is a deterministic-time identity

Weak Markov property of Brownian motion

For every bounded measurable function f and $h \geq 0$,

$$\boxed{\mathbb{E}[f(W_{t+h}) \mid \mathcal{F}_t] = \mathbb{E}[f(W_{t+h}) \mid W_t] = (P_h f)(W_t)}.$$

Proof. Write

$$W_{t+h} = W_t + \widetilde{W}_h, \qquad \widetilde{W}_h \sim \mathcal{N}(0, h), \qquad \widetilde{W}_h \perp\!\!\!\perp \mathcal{F}_t.$$

Conditioning on $\mathcal{F}_t$, only W_t remains random through the translation:

$$(P_h f)(x) = \int_{\mathbb{R}} f(x+y) \frac{e^{-y^2/(2h)}}{\sqrt{2\pi h}}\, dy.$$

Weak Markov exercise: forecast from the current state

At time 2, the whole path $(W_u)_{0\le u\le 2}$ has been observed. On the branch where $W_2 = 1$:

Your turn – 2 minutes

Without using the detailed past path, compute

1. $\mathbb{P}(W_5 > 2 \mid W_2 = 1)$;
2. $\mathbb{E}[W_5^2 \mid \mathcal{F}_2]$, then evaluate it when $W_2 = 1$;
3. the conditional law of W_5 given $\mathcal{F}_2$.

Diagnostic question

Would knowing that the path crossed zero three times before time 2 change any answer once W_2 is known?

Weak Markov debrief: translate by the observed state

Write

$$W_5 = W_2 + (W_5 - W_2) = W_2 + \sqrt{3}\,Z, \qquad Z \sim \mathcal{N}(0,1), \quad Z \perp\!\!\!\perp \mathcal{F}_2.$$

Therefore

$$W_5 \mid \mathcal{F}_2 \sim \mathcal{N}(W_2, 3), \qquad \mathbb{E}[W_5^2 \mid \mathcal{F}_2] = W_2^2 + 3.$$

On the branch $W_2 = 1$,

$$\mathbb{P}(W_5 > 2 \mid W_2 = 1) = 1 - \Phi\left(\frac{1}{\sqrt{3}}\right) \approx 0.282, \qquad \mathbb{E}[W_5^2 \mid W_2 = 1] = 4.$$

Takeaway

At a deterministic time, the present level is sufficient; the route taken to reach it is irrelevant for the future conditional law.

Brownian motion is also a martingale

For $0 \leq s < t$,

$$\mathbb{E}[W_t \mid \mathcal{F}_s] = \mathbb{E}[W_s + (W_t - W_s) \mid \mathcal{F}_s] = W_s.$$

Martingale interpretation

Given all currently available information, the conditional expected future value equals the current value.

- Markov: the current state summarizes the relevant past.
- Martingale: the current state is the best conditional mean forecast.

The two concepts are related here but are not synonymous.

Stopping Times and Strong Markov Property

Discrete time makes the no-look-ahead rule visible

Let $\mathcal{F}_n = \sigma(X_0, \ldots, X_n)$ be the information observed by date n.

Discrete stopping time

An integer-valued random time τ is a stopping time if

$$\{\tau \le n\} \in \mathcal{F}_n \qquad \text{for every } n.$$

Observable: first hit

$$\tau_a = \inf\{n : X_n \ge a\}.$$

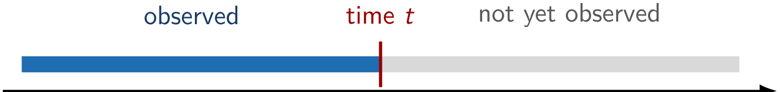
At date n, the observed path reveals whether a hit occurred.

Anticipative: last zero

$$\rho = \sup\{n \le N : X_n = 0\}.$$

Before N, a later return to zero may still occur.

A stopping rule must be executable with information available now



Operational question

At time t, can I determine whether the rule has already triggered?

- If yes for every t, the random time is admissible.
- If the answer requires seeing the future path, it is not.

The definition turns the operational question into measurability

Let $(\mathcal{F}_t)_{t \geq 0}$ represent the information observed through time.

Stopping time

A random time $\tau : \Omega \to [0, \infty]$ is a stopping time if

$$\{\tau \leq t\} \in \mathcal{F}_t \qquad \text{for every } t \geq 0.$$

The event $\{\tau \leq t\}$ means "the rule has triggered by time t." It must be answerable using only information in $\mathcal{F}_t$.

Important distinction

A stopping time is random, but it is not anticipative.

Deterministic times and first hitting times are stopping times

Deterministic time

For $\tau = t_0$,

$$\{\tau \le t\} = \begin{cases} \emptyset, & t < t_0, \\ \Omega, & t \ge t_0. \end{cases}$$

First hit of level a

$$\tau_a = \inf\{t \ge 0 : W_t = a\}.$$

By continuity,

$$\{\tau_a \le t\} = \left\{ \max_{0 \le u \le t} W_u \ge a \right\} \in \mathcal{F}_t$$

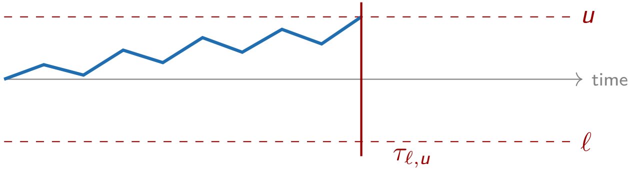
for $a > 0$.

Both rules can be verified as the path unfolds.

First exit from an interval uses only the observed path

For $\ell < u$, define

$$\tau_{\ell,u} = \inf\{t \geq 0 : W_t \notin (\ell, u)\}.$$

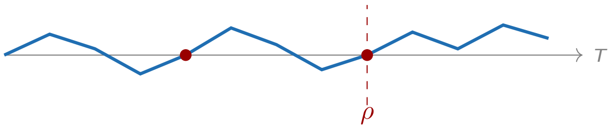


By time t, we know whether the observed path has left the interval. No future information is needed.

The last zero before a horizon is not a stopping time

Fix $T > 0$ and define

$$\rho = \sup\{0 \leq u \leq T : W_u = 0\}.$$



When the path is at zero at time $t < T$, we cannot know whether it will return to zero later. Identifying the *last* zero requires the future path.

Quick check: which random times are stopping times?

Fix $T > 0$. Consider:

1. $\tau_1 = \inf\{t \geq 0 : W_t \geq 1\}$;
2. $\tau_2 = T/2$;
3. $\tau_3 = \inf\{t \geq T/2 : W_t = 0\}$;
4. $\tau_4 = \arg\max_{0 \leq t \leq T} W_t$.

Quick check: which random times are stopping times?

Fix $T > 0$. Consider:

1. $\tau_1 = \inf\{t \geq 0 : W_t \geq 1\}$;
2. $\tau_2 = T/2$;
3. $\tau_3 = \inf\{t \geq T/2 : W_t = 0\}$;
4. $\tau_4 = \arg\max_{0 \leq t \leq T} W_t$.

Answer

τ_1, τ_2, τ_3 are stopping times. The time τ_4 of the maximum is not: deciding that the current value is the maximum requires observing the rest of the path.

$\mathcal{F}_\tau$ is the information available when the rule triggers

Stopped sigma-field

For a stopping time τ, define

$$\mathcal{F}_\tau = \Big\{ A \in \mathcal{F} : \; A \cap \{\tau \le t\} \in \mathcal{F}_t \text{ for every } t \ge 0 \Big\}.$$

An event in $\mathcal{F}_\tau$ can be decided using only the information revealed by the random time τ.

Discrete-time picture

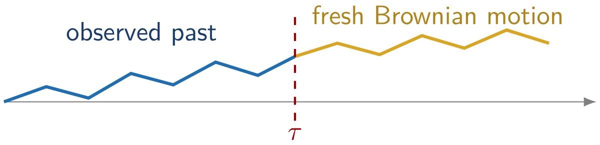
If τ takes integer values, then $A \in \mathcal{F}_\tau$ iff $A \cap \{\tau = n\} \in \mathcal{F}_n$ for every n. On each branch $\{\tau = n\}$, only observations through date n may be used.

Strong Markov means that Brownian motion restarts at a stopping time

Let τ be an almost surely finite stopping time. Then

$$\widetilde{W}_u = W_{\tau+u} - W_\tau, \qquad u \geq 0,$$

is a Brownian motion independent of the information $\mathcal{F}_\tau$ available when stopping.



Ordinary Markov restarts at a fixed time; strong Markov restarts when an observable event occurs.

Strong Markov exercise: restart when a barrier is reached

Let

$$\tau_1 = \inf\{t \geq 0 : W_t = 1\}.$$

Brownian motion reaches level 1 almost surely, so $W_{\tau_1} = 1$.

Your turn – 3 minutes

Use strong Markov to answer:

1. What is the conditional law of W_{τ_1+h} given $\mathcal{F}_{\tau_1}$?
2. Compute $\mathbb{P}(W_{\tau_1+2} \geq 2 \mid \mathcal{F}_{\tau_1})$.
3. If $\rho = \inf\{u \geq 0 : W_{\tau_1+u} = 3\}$, which familiar first-passage time has the same law as ρ?

The key is to recenter both time and space at the random point $(\tau_1, 1)$.

Strong Markov debrief: the post-hit increment is fresh

Define $B_h = W_{\tau_1+h} - W_{\tau_1}$. Strong Markov gives

$$(B_h)_{h\geq 0} \text{ is Brownian and independent of } \mathcal{F}_{\tau_1}.$$

Hence

$$W_{\tau_1+h} \mid \mathcal{F}_{\tau_1} \sim \mathcal{N}(1, h)$$

and

$$\mathbb{P}(W_{\tau_1+2} \geq 2 \mid \mathcal{F}_{\tau_1}) = 1 - \Phi\left(\frac{1}{\sqrt{2}}\right) \approx 0.240.$$

To reach level 3 after τ_1, the fresh Brownian motion B must travel from 0 to 2. Therefore

$$\rho \overset{d}{=} \tau_2.$$

Contrast

Weak Markov restarts at a fixed clock time; strong Markov restarts at an observable random time.

Strong Markov is the bridge from stopping to reflection

Suppose a Brownian path first hits level $a > 0$ at

$$\tau_a = \inf\{u \geq 0 : W_u = a\}.$$

After τ_a,

$$W_{\tau_a+u} - a$$

is a fresh Brownian motion, independent of how the path reached a.

Why this matters

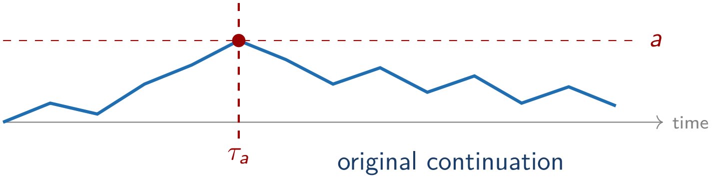
Because a fresh Brownian motion is symmetric, we may flip the sign of every increment after τ_a without changing its law.

That sign flip is exactly the reflection principle.

Reflection Principle and Hitting Probabilities

Step 1: stop at the first barrier hit

Let $a > 0$ and $\tau_a = \inf\{u \geq 0 : W_u = a\}$.

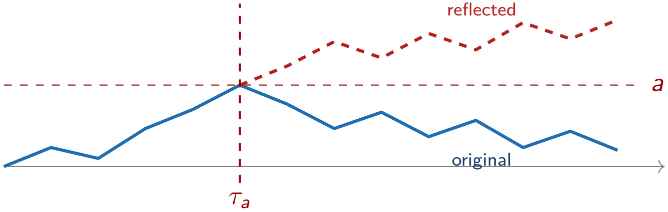


The first part of the path identifies when reflection starts. Only the continuation after τ_a will change.

Step 2: reflect only the fresh continuation

For a path that hits a, define

$$W_u^\star = \begin{cases} W_u, & u \le \tau_a, \\ 2a - W_u, & u > \tau_a. \end{cases}$$



Strong Markov plus symmetry implies that the reflected path has the same Brownian probability law.

Step 3: pair terminal events path by path

Let

$$M_t = \max_{0 \le u \le t} W_u.$$

Among paths that reach a before t, reflection gives a one-to-one correspondence:

$$\{M_t \ge a,\ W_t < a\} \quad \longleftrightarrow \quad \{W_t > a\}.$$

$$\boxed{\text{hit } a\text{, finish below } a} \xleftrightarrow{\text{reflect}} \boxed{\text{finish above } a}$$

The equality of probabilities follows from this measure-preserving pairing.

Theorem: reflection doubles a Gaussian tail

The event $\{M_t \geq a\}$ splits into two disjoint cases:

$$\{M_t \geq a\} = \{W_t \geq a\} \cup \{M_t \geq a,\ W_t < a\}.$$

Reflection gives

$$\mathbb{P}(M_t \geq a,\ W_t < a) = \mathbb{P}(W_t > a).$$

Since $\mathbb{P}(W_t = a) = 0$,

$$\boxed{\mathbb{P}(M_t \geq a) = 2\mathbb{P}(W_t \geq a)} = 2\left(1 - \Phi\left(\frac{a}{\sqrt{t}}\right)\right).$$

The running maximum has the same one-dimensional distribution as $|W_t|$.

Corollary: first-passage times have an explicit law

For $\tau_a = \inf\{u \geq 0 : W_u = a\}$,
$$\{\tau_a \leq t\} = \{M_t \geq a\}.$$

Therefore
$$\boxed{\mathbb{P}(\tau_a \leq t) = 2\left(1 - \Phi\left(\frac{a}{\sqrt{t}}\right)\right)}.$$

Differentiating yields
$$f_{\tau_a}(t) = \frac{a}{\sqrt{2\pi t^3}} \exp\left(-\frac{a^2}{2t}\right), \qquad t > 0.$$

Thus Brownian motion eventually reaches every fixed level a with probability one.

Application: a four-year barrier probability

What is the probability that Brownian motion reaches level 1 before time 4?

Since $\{\tau_1 \leq 4\} = \{M_4 \geq 1\}$, the reflection principle gives

$$\mathbb{P}(\tau_1 \leq 4) = 2\left(1 - \Phi\left(\frac{1}{\sqrt{4}}\right)\right) = 2\big(1 - \Phi(0.5)\big) \approx 0.617.$$

Interpretation

Although $W_4 \sim \mathcal{N}(0, 4)$, the barrier event depends on the whole path, not only on the terminal value. Reflection converts that path-dependent event into a Gaussian tail.

Finance application: a barrier claim is a first-passage problem

Under the risk-neutral measure,

$$S_t = S_0 \exp(\nu t + \sigma W_t)\,, \qquad \nu = r - \tfrac{1}{2}\sigma^2.$$

For an upper barrier $H > S_0$, define $\tau_H = \inf\{t : S_t \geq H\}$ and $a = \log(H/S_0)$. Then

$$\{\tau_H \leq T\} = \left\{ \max_{0\leq t\leq T} (\nu t + \sigma W_t) \geq a \right\}.$$

Zero log-drift case $\nu = 0$

The ordinary reflection principle gives

$$\boxed{\mathbb{P}(\tau_H \leq T) = 2\left[1 - \Phi\left(\frac{\log(H/S_0)}{\sigma\sqrt{T}}\right)\right].}$$

With nonzero ν, the drifted reflection formula adds an exponential tilt.

Worked finance example: value a one-touch digital

Consider a one-year contract with

$$S_0 = 100, \qquad H = 120, \qquad T = 1, \qquad r = 2\%, \qquad \sigma = 20\%.$$

Here $\nu = r - \frac{1}{2}\sigma^2 = 0$, and therefore

$$a = \log(1.2) \approx 0.1823, \qquad \frac{a}{\sigma\sqrt{T}} \approx 0.9116.$$

Reflection gives the probability of touching 120 before maturity:

$$\mathbb{P}(\tau_{120} \leq 1) = 2\big[1 - \Phi(0.9116)\big] \approx 0.362.$$

A claim paying one dollar at T after a touch is worth

$$V_0 = e^{-rT}\mathbb{P}(\tau_{120} \leq 1) \approx e^{-0.02}(0.362) \approx 0.355.$$

Financial interpretation. Reflection turns continuous monitoring into a Gaussian-tail calculation and underlies barrier-option formulas.

Quadratic Variation and Brownian Roughness

Non-differentiability suggests a different notion of variation

On a partition $0 = t_0 < \cdots < t_n = T$, write

$$\Delta_i W = W_{t_{i+1}} - W_{t_i}.$$

For an evenly spaced grid, $\Delta_i W$ typically has size $\sqrt{T/n}$. Thus

$$\sum_{i=0}^{n-1} (\Delta_i W)^2 \quad \text{has typical size} \quad n \times \frac{T}{n} = T.$$

Contrast with smooth paths

For a differentiable path, increments have size T/n, so the sum of squared increments has size $n(T/n)^2 \to 0$.

Brownian paths have infinite ordinary variation, but their squared increments accumulate to a finite limit. This is the useful replacement for a nonexistent derivative.

Quadratic variation records accumulated squared increments

For a continuous process X and partitions

$$\pi_n = \{0 = t_0^n < \cdots < t_{k_n}^n = t\}, \qquad |\pi_n| \to 0,$$

define

$$[X]_t = \lim_{n\to\infty} \sum_i \left(X_{t_{i+1}^n} - X_{t_i^n} \right)^2,$$

when the limit exists in probability.

Interpretation

$[X]_t$ measures the amount of fine-scale path fluctuation accumulated up to time t.

Quadratic variation is itself an accumulated-variation process

When it exists, $([X]_t)_{t\geq 0}$ starts at zero and is nondecreasing. It measures fluctuation, not the direction of moves.

Elementary properties

If A is continuous with finite variation, then

$$[A]_t = 0, \qquad [X+A]_t = [X]_t, \qquad [aX]_t = a^2[X]_t.$$

Proof idea: fine increments of A are uniformly small, while their absolute sum remains bounded; hence their squared sum and cross-term with X vanish.

Numerical meaning

On a fine observation grid, the realized sum $\sum_i (\Delta_i X)^2$ estimates the accumulated quadratic variation.

Smooth paths have zero quadratic variation

Let x be continuously differentiable on $[0, T]$. By the mean value theorem,

$$x_{t_{i+1}} - x_{t_i} = x'(\xi_i)(t_{i+1} - t_i).$$

Hence

$$\sum_i (x_{t_{i+1}} - x_{t_i})^2 \le \|x'\|_\infty^2 |\pi_n| \sum_i (t_{i+1} - t_i) = \|x'\|_\infty^2 T |\pi_n| \to 0.$$

Example

For $x_t = t$, an equal grid gives $n(T/n)^2 = T^2/n \to 0$.

Brownian motion has quadratic variation equal to elapsed time

Fundamental result

For Brownian motion and any fixed $t \geq 0$,

$$\boxed{[W]_t = t}$$

in probability along partitions whose mesh tends to zero.

On an equal grid with $\Delta t = t/n$,

$$\mathbb{E}\left[\sum_{i=0}^{n-1} (\Delta_i W)^2\right] = \sum_{i=0}^{n-1} \mathbb{E}[(\Delta_i W)^2] = n\Delta t = t.$$

The randomness of the sum vanishes as the grid is refined.

A short variance calculation explains concentration around t

For independent $\Delta_i W \sim \mathcal{N}(0, \Delta t)$,

$$\operatorname{Var}\big((\Delta_i W)^2\big) = 2(\Delta t)^2.$$

Therefore

$$\operatorname{Var}\left(\sum_{i=0}^{n-1} (\Delta_i W)^2\right) = 2n(\Delta t)^2 = \frac{2t^2}{n} \longrightarrow 0.$$

Thus the sum of squared increments converges to t in L^2, hence in probability.

Meaning

Each squared increment is random, but their accumulated total becomes deterministic.

What $[X]_t$ tells us about a process

Process	Quadratic variation	What survives at fine scale
smooth or finite-variation A	$[A]_t = 0$	no rough component
Brownian motion W	$[W]_t = t$	one variance unit per time unit
$X_t = A_t + \sigma W_t$	$[X]_t = \sigma^2 t$	volatility, not drift

Key message

Quadratic variation is a pathwise record of accumulated volatility. It ignores smooth drift and retains persistent small-scale randomness.

This is why high-frequency squared returns are the starting point for realized-volatility estimation.

The bracket corrects the square of a martingale

Quadratic-variation martingale

For a continuous square-integrable martingale X, there is a unique increasing predictable process $\langle X \rangle$, starting from zero, such that

$$\boxed{X_t^2 - \langle X \rangle_t \text{ is a martingale.}}$$

For continuous martingales, $\langle X \rangle = [X]$.

For Brownian motion, $\langle W \rangle_t = [W]_t = t$. Since

$$\mathbb{E}[W_t^2 \mid \mathcal{F}_s] = W_s^2 + (t - s).$$

we obtain

$$\mathbb{E}[W_t^2 - t \mid \mathcal{F}_s] = W_s^2 - s.$$

Hence $W_t^2 - t$ is a martingale.

The Itô multiplication table summarizes the new calculus

Brownian scaling suggests dW_t has size $\sqrt{dt}$. Hence

$$(dW_t)^2 = dt, \qquad dW_t\,dt = 0, \qquad (dt)^2 = 0.$$

$\times$	dt	dW_t
dt	0	0
dW_t	0	dt

Preview

The identity $(dW_t)^2 = dt$ is why Itô's formula contains a second-derivative term.

Discrete trading gains lead naturally to stochastic integrals

On a trading grid,

$$G_T^{(n)} = \sum_{i=0}^{n-1} \Delta_{t_i} \left(S_{t_{i+1}} - S_{t_i} \right),$$

where Δ_{t_i} is chosen using information available at t_i. As the mesh tends to zero, the limiting gain is written

$$G_T = \int_0^T \Delta_t \, dS_t.$$

- Adaptedness prevents look-ahead trading.
- Quadratic variation determines the calculus of the limit.
- The next lectures make this construction and Itô's formula precise.

From Brownian Motion to a Positive Price Model

An additive Brownian price can become negative

The model

$$S_t = S_0 + \mu t + \sigma W_t$$

has Gaussian values with support on all of $\mathbb{R}$.

- Negative prices are possible.
- Absolute volatility is constant, regardless of the price level.
- Financial returns are usually modeled multiplicatively instead.

Exponentiating Brownian log-returns keeps prices positive

Assume the log-price follows

$$\log \frac{S_t}{S_0} = \left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma W_t.$$

Then

$$\boxed{S_t = S_0 \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma W_t\right)}$$

is a geometric Brownian motion.

- $S_t > 0$ for every t.
- Log-returns over disjoint intervals are independent.
- Their variance is proportional to elapsed time.

Worked example: one-year distribution under GBM

Let

$$S_0 = 100, \qquad \mu = 8\%, \qquad \sigma = 20\%, \qquad t = 1.$$

Then

$$\log \frac{S_1}{100} \sim \mathcal{N}(0.08 - \tfrac{1}{2}(0.20)^2,\ (0.20)^2) = \mathcal{N}(0.06, 0.04).$$

Therefore

$$\text{Median}(S_1) = 100e^{0.06} \approx 106.18, \qquad \mathbb{E}[S_1] = 100e^{0.08} \approx 108.33.$$

The mean exceeds the median because a lognormal distribution is right-skewed.

The drift controls the mean; volatility controls dispersion

For geometric Brownian motion,

$$\mathbb{E}[S_t] = S_0 e^{\mu t}, \qquad \mathrm{Var}(S_t) = S_0^2 e^{2\mu t}\left(e^{\sigma^2 t} - 1\right).$$

Over a short interval Δt,

$$\log \frac{S_{t+\Delta t}}{S_t} \approx \mathcal{N}\left(\left(\mu - \tfrac{1}{2}\sigma^2\right)\Delta t,\, \sigma^2 \Delta t\right).$$

Square-root-of-time rule

The standard deviation of a log-return over Δt is $\sigma\sqrt{\Delta t}$.

Quadratic variation identifies accumulated log-price volatility

Let

$$X_t = \log S_t = \log S_0 + \left(\mu - \tfrac{1}{2}\sigma^2\right) t + \sigma W_t.$$

The finite-variation drift has zero quadratic variation, so

$$\boxed{[X]_t = \sigma^2 [W]_t = \sigma^2 t.}$$

Interpretation

Quadratic variation filters out the smooth drift and retains the cumulative variance generated by Brownian shocks.

This connection underlies realized-volatility estimators based on high-frequency squared returns.

GBM is useful, but its assumptions are visible

What it captures

- positive prices;
- random continuous returns;
- explicit conditional laws;
- a tractable volatility parameter.

What it misses

- jumps and extreme tails;
- changing volatility;
- volatility smiles;
- long-range market effects.

Modeling lesson

Brownian motion is a foundational building block, not a complete description of every market feature.